ANALYSIS REPORT

Wave 3 Harmonic-Prior Residual Campaign Results Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

The first real `Wave 3` harmonic-prior residual campaign completed all six runs successfully; the scalar winner is the backward pointwise-control branch, but the campaign does not replace the current program winner.

Campaign Artifacts

Artifact	Path
Campaign output	`output/training_campaigns/2026-06-15-14-01-15_wave3_harmonic_prior_residual_campaign_2026_06_14`
Execution report	`output/training_campaigns/2026-06-15-14-01-15_wave3_harmonic_prior_residual_campaign_2026_06_14/campaign_execution_report.md`
Leaderboard	`output/training_campaigns/2026-06-15-14-01-15_wave3_harmonic_prior_residual_campaign_2026_06_14/campaign_leaderboard.yaml`
Best run pointer	`output/training_campaigns/2026-06-15-14-01-15_wave3_harmonic_prior_residual_campaign_2026_06_14/campaign_best_run.yaml`
Manifest	`output/training_campaigns/2026-06-15-14-01-15_wave3_harmonic_prior_residual_campaign_2026_06_14/campaign_manifest.yaml`
Planning report	`doc/reports/campaign_plans/wave3_wave4/2026-06-14-19-54-55_wave3_harmonic_prior_residual_campaign_plan_report.md`
Technical package	`doc/technical/2026-06/2026-06-14/2026-06-14-19-54-55_wave3_harmonic_prior_residual_campaign_package.md`

Execution Summary

Field	Value
Campaign	`wave3_harmonic_prior_residual_campaign_2026_06_14`
Generated at	`2026-06-15T15:30:20`
Completed runs	`6`
Failed runs	`0`
Profiles	`pointwise_control` , `smooth_l1_structured`
Surfaces	`global` , `Fw` , `Bw`
Best scalar run	`te_wave3_harmonic_prior_residual_pointwise_control_bw`
Best scalar family	`wave3_harmonic_prior_residual_pointwise_control_bw`
Best scalar test MAE	`0.003363`

Field	Value
Best scalar test RMSE	`0.003902`
Program winner changed	`no`
Current program winner	`te_periodic_gru_sequence_remote_Bw`
Track 2 status	not run during normal closeout

Directional Branch Results

Surface	Candidate	Profile	Test MAE	Test RMSE	Val MAE
`global`	`sl1_global`	`sl1`	0.003403	0.003785	0.003633
`Fw`	`pw_fw`	`pw`	0.003382	0.003779	0.003315
`Bw`	`pw_bw`	`pw`	0.003363	0.003902	0.003634

Wave 3 Leaderboard

Candidate labels are aliases for the corresponding `te_wave3_harmonic_prior_residual_*` run names stored in the campaign leaderboard. All candidates have `7,283` trainable parameters.

Rank	Candidate	Surface	Test MAE	Test RMSE	Val MAE
1	`pw_bw`	`Bw`	0.003363	0.003902	0.003634
2	`pw_fw`	`Fw`	0.003382	0.003779	0.003315
3	`sl1_global`	`global`	0.003403	0.003785	0.003633
4	`sl1_bw`	`Bw`	0.003431	0.003953	0.003644
5	`pw_global`	`global`	0.003451	0.003851	0.003611
6	`sl1_fw`	`Fw`	0.003527	0.003900	0.003310

Profile Comparison

Surface	Pointwise MAE	Smooth-L1 MAE	Better Profile	Relative Gap
`global`	0.003451	0.003403	`sl1`	1.40%
`Fw`	0.003382	0.003527	`pw`	4.12%
`Bw`	0.003363	0.003431	`pw`	1.97%

Interpretation

`Wave 3` validates the lightweight harmonic-prior residual implementation as a real trainable branch, with all six directional candidates completing cleanly and registering as implemented benchmarks. The best scalar result is backward only, while the best global result comes from the `smooth_l1_structured` profile.

The result is useful but not yet promotable. The best `Wave 3` scalar test MAE of `0.003363 deg` is weaker than the current program winner `te_periodic_gru_sequence_remote_Bw` at `0.002344 deg`, and also weaker than the best recent `Track 2H` mixture-density backward branch on scalar MAE. The main positive signal is architectural: the candidate has only `7,283` trainable parameters and keeps an explicit harmonic-prior residual structure.

The profile comparison does not support treating the robust `SmoothL1` branch as a universal default. It helps on the global surface, but the plain pointwise-control profile wins on both direction-specific surfaces. The next decision should therefore be curve-first and direction-specific, not based on scalar training MAE alone.

Registry Effects

Registry Scope	Effect
Family registries	Six new `wave3_harmonic_prior_residual_*` family bests are registered.
Program registry	No program-best promotion; `te_periodic_gru_sequence_remote_Bw` remains current.
Active campaign state	Cleared by this closeout after report and PDF validation.
Master summary	Updated to show `Wave 3` campaign closeout and separate `Track 2` boundary.

Track 2 Boundary

Normal campaign closeout intentionally did not execute the heavy official `Track 2` offline verification matrix. The next optional step is a separate `Track 2` verification refresh for the six `Wave 3` candidates, including collage and overlay reports, so the harmonic-prior residual branch can be compared on curve shape, offset, and direction-specific compensation behavior.

Closeout Decision

Accept the campaign as successfully closed at the training-campaign level. Do not promote a new program winner from scalar metrics. Plan official `Track 2` verification as the next separate acceptance step before deciding whether `Wave 3` should feed the later multi-head or `Wave 4` integration roadmap.